The Four Arches

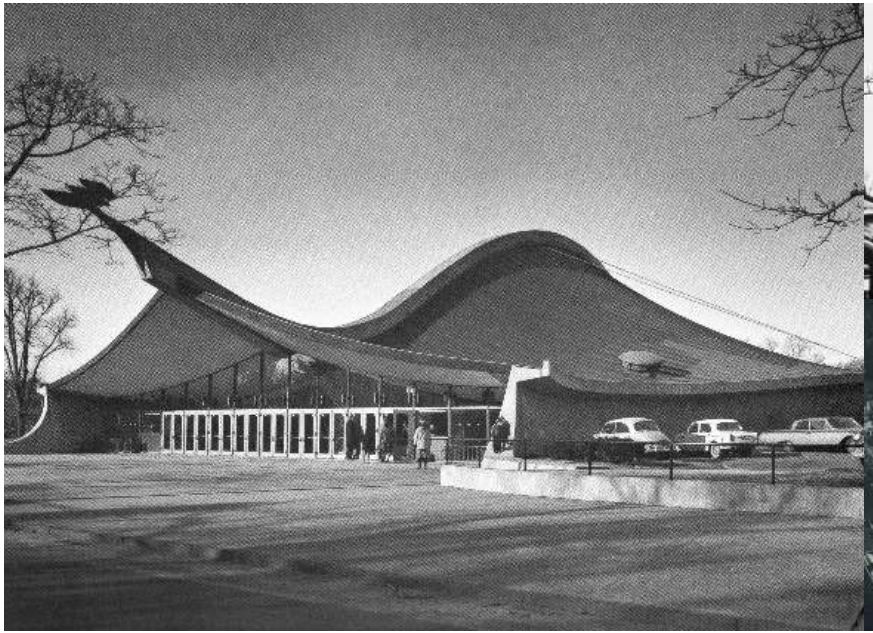
at Les Tres Xemeneies

Tower Design Responding to Global Rising Sea Levels & the Modern Housing Crisis

Final Presentation | Ryan Guerra, Luc Carner, Kai Aguilar Udall, Teague Roark

Investigation & Exploration - Precedent Analysis

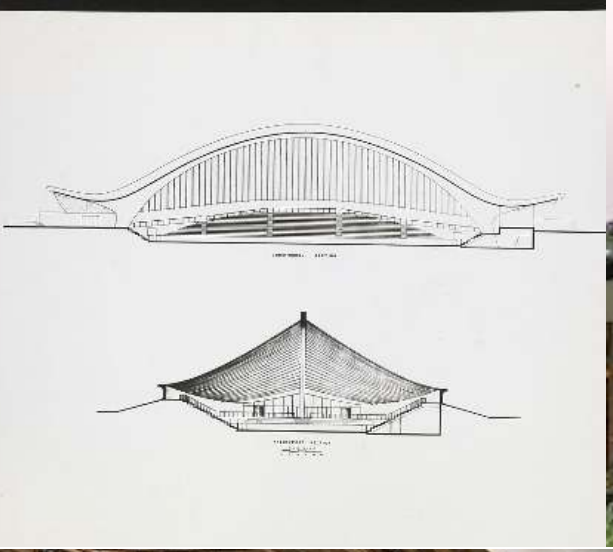
David S. Ingalls Rink - Yale University, New Haven, Connecticut, U.S.A. - Eero Saarinen - 1958









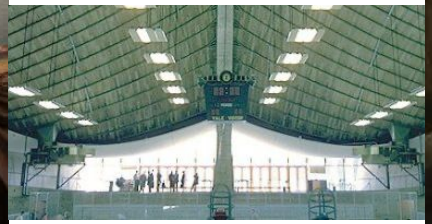







YALE HOCKEY







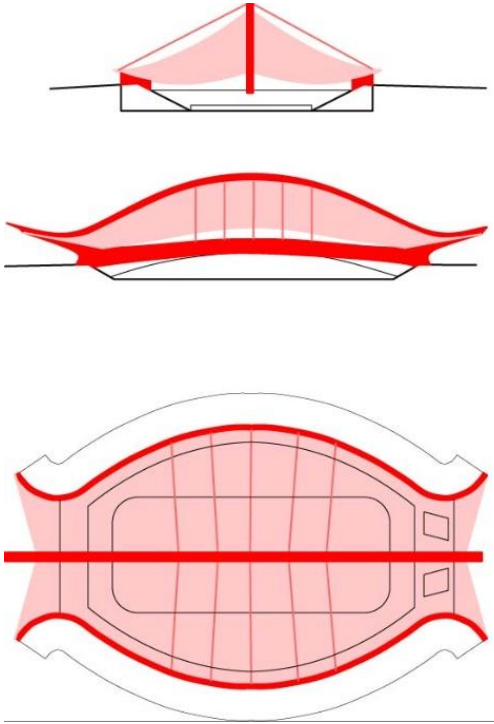
Investigation & Exploration - Structure

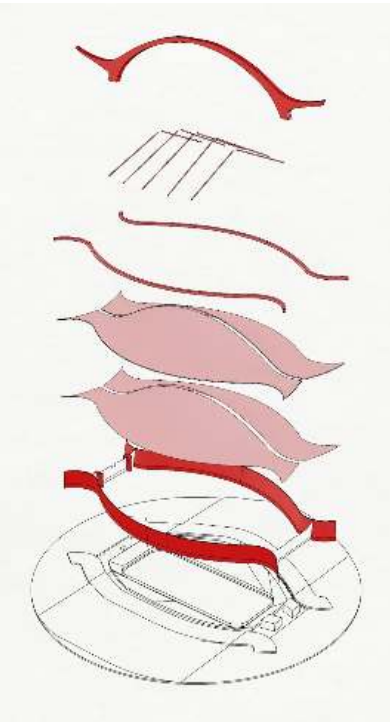
Geometry is derived directly from **structural necessity**

Primary Structure: concrete **catenary arch** and load-bearing walls

Secondary Structure: **cable net system** distributing loads

Tertiary Structure: wooden and aluminum **roofing**





Investigation & Exploration - Spatial Hierarchy

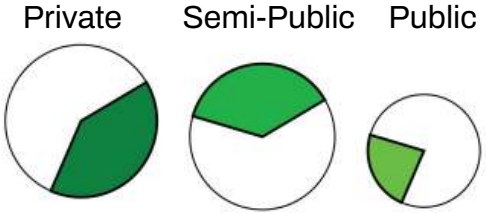
Restricted spaces occupy the largest area.

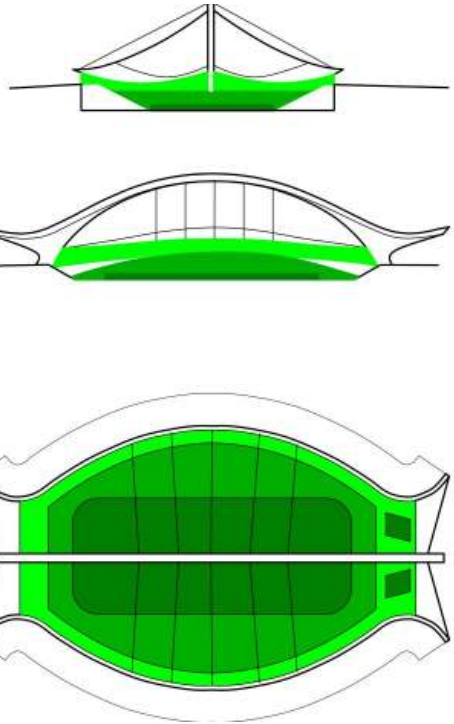
Spaces remain compressed but continuous.

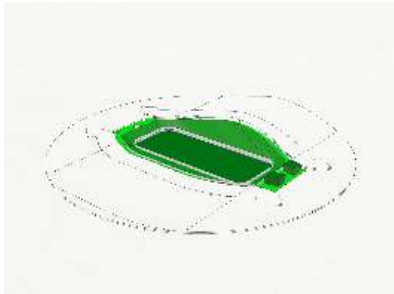
Public Spaces - 24.5% of Interior Space (10,025 SqFt)

Semi-Public Spaces - 36.5% of Interior Space (14,948 SqFt)

Restricted Spaces - 39% of Interior Space (15,934 SqFt)



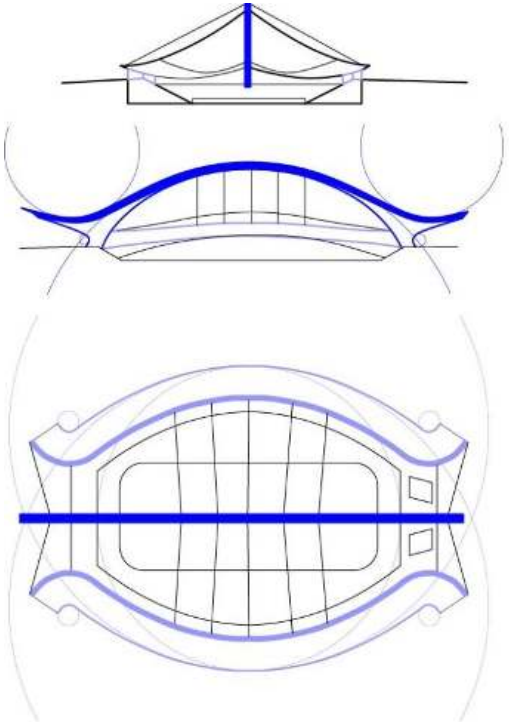


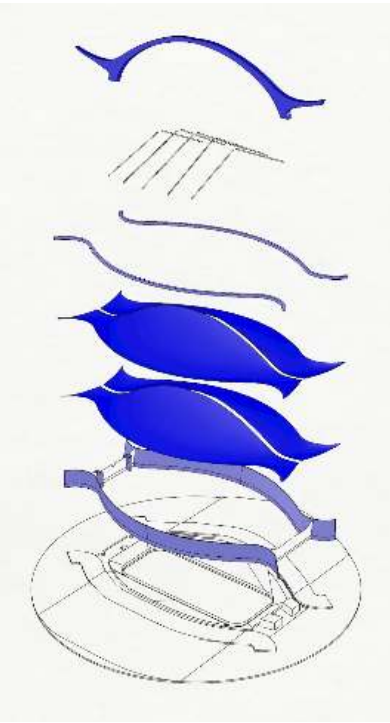


Investigation & Exploration - Curvature

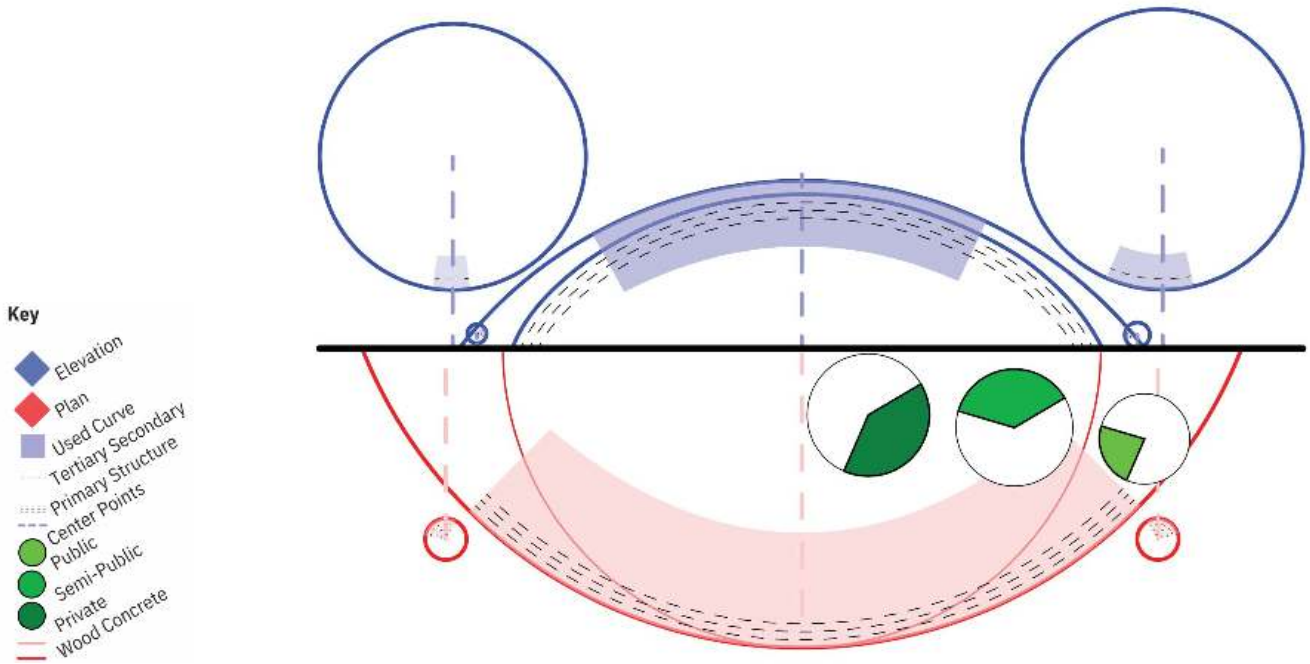
Repetition of **curved geometry** relating back to the structure in elevation and plan.

Main catenary defines general form of the building, creating a theme for the rest of the form to follow.





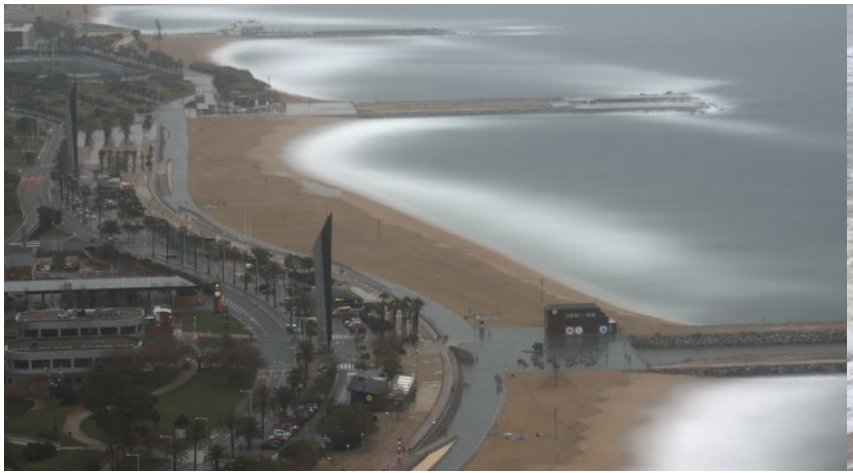
Investigation & Exploration - Combined Deconstruction Diagram



Speculative Framework - Current Condition & Problem

Beach erosion

- The Catalan coast **rises by an approximate 3 millimeters** per year.
- The coast is also eroding, **losing about 30,000 m³** due to intense storms and the rise in sea levels.
- With shorter beaches and strong waves forms storms the water can heavily **damage the surrounding infrastructure**.
- It is quite **costly for the city** as they have to refill the beaches with imported sand for tourist season







Speculative Framework - 100 Year Projection

If no action is taken, Barcelona’s coastline will continue to retreat, flooding beaches, damaging infrastructure, and pushing public life inland. The city risks losing its primary open edge as rising seas and storm surges erase the beach as a civic space.





Site Analysis - Site Context, Sant Adrià de Besòs

Les Tres Xemeneies, Sant Adria de Besos, about **6 km northeast of central Barcelona**.
It occupies a transitional zone between:

- The Mediterranean coast
- The lower basin of the Besòs River
- Urban continuum connecting Barcelona with Badalona

Key metropolitan infrastructures nearby:

- Las Tres Chimeneas de Barcelona
- Coastal railway corridor
- C-31 highway
- Coastal energy infrastructure and industrial zones



Site Analysis - Proposed Projects

Besòs Waterfront Transformation

• 32-hectare redevelopment of former industrial waterfront

• New tram infrastructure and pedestrian connections improve regional accessibility

• Public parks and housing reconnect the city to the Mediterranean coast

Highlights the need for projects that support public access, ecological restoration, and transit programming.



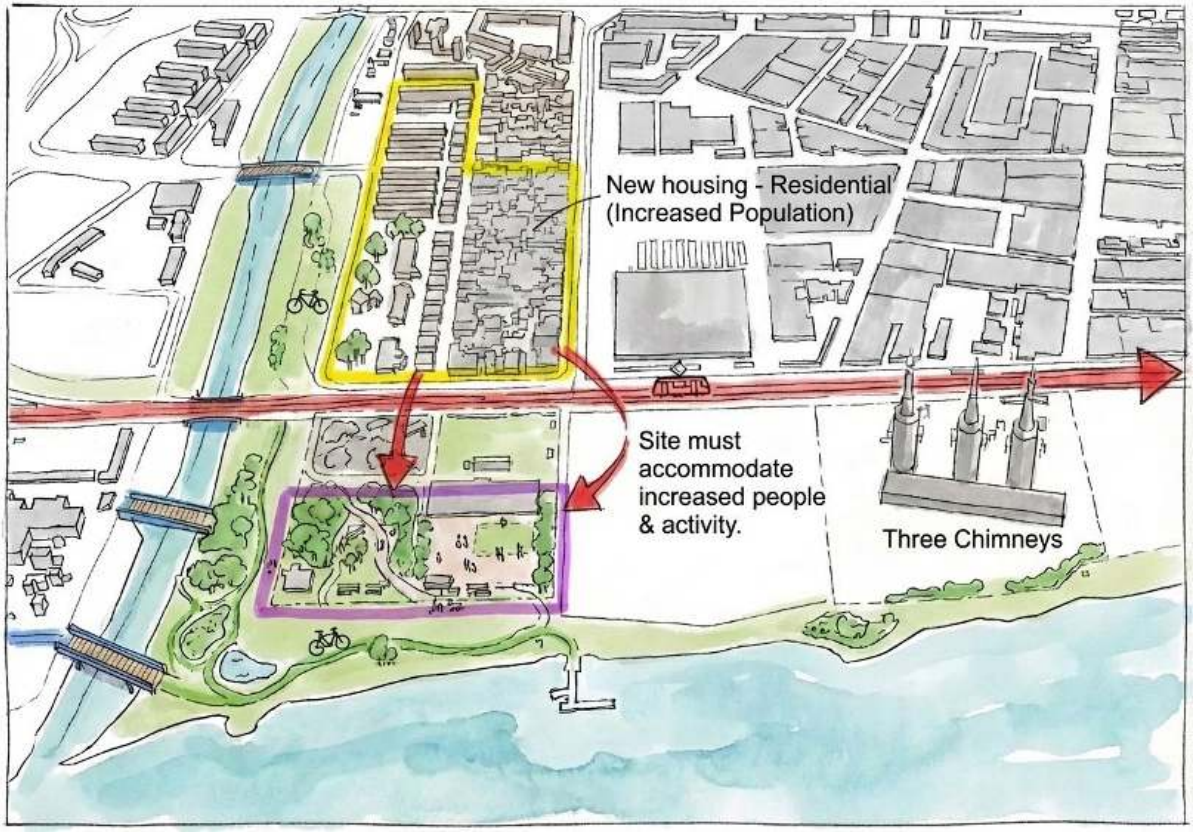
Site Analysis - Human Needs Around the New District

Analyze the social infrastructure needed for the new district surrounding the Three Chimneys development.

The redevelopment plan includes:

- ~1,700–1,800 new housing units
- a large waterfront park
- the Catalunya Media City audiovisual hub - Three Chimneys

This means the area will transition from an industrial zone to a dense mixed-use neighborhood.



Site Analysis - Circulation + Functional Urban Node

How a new building should function within the future transportation and public circulation network of the Tres Xemeneies district.

The urban redevelopment will include:

- a 1.4 km tram extension
- new pedestrian bridge
- bike and pedestrian routes
- redesigned streets and parks

These elements aim to reconnect the city, river, and sea.

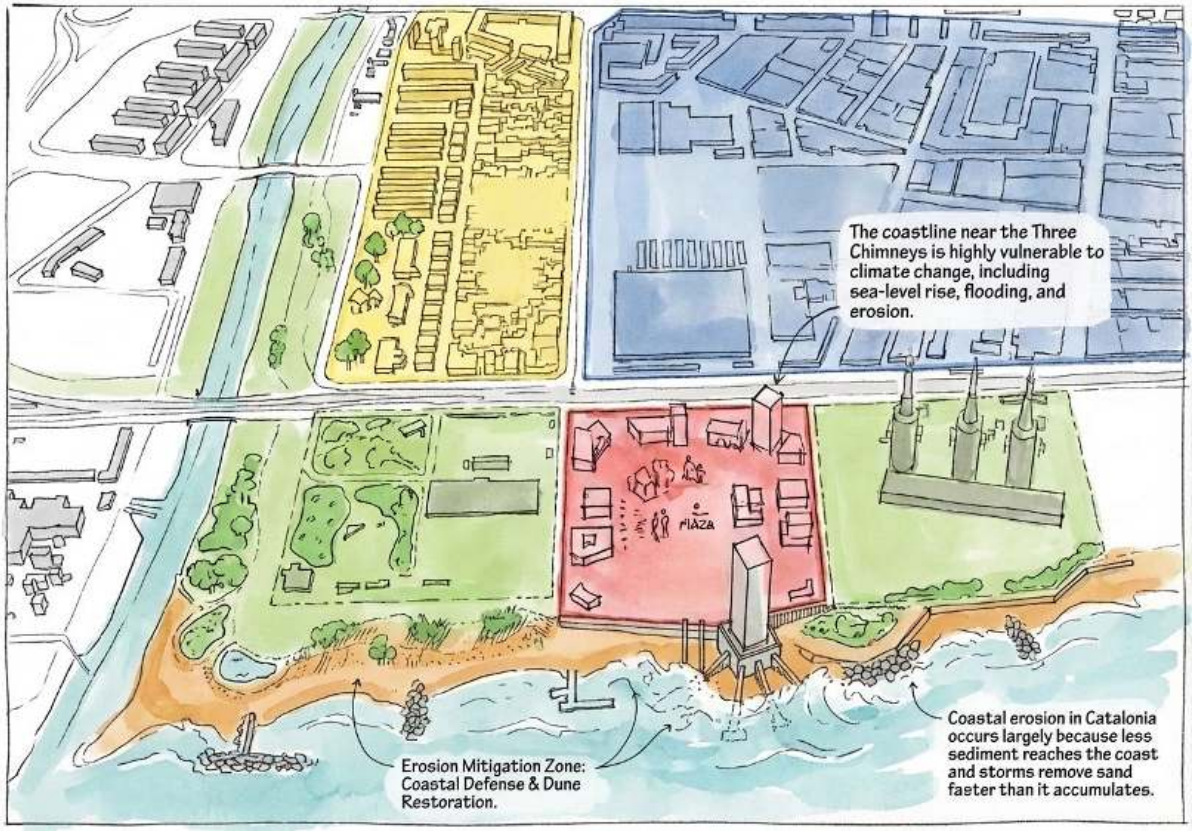


Site Analysis - Coastal Erosion & Beach Resilience

Utilize soft shoreline defenses like restored dunes and native vegetation to hold sediment in place and absorb storm wave energy.

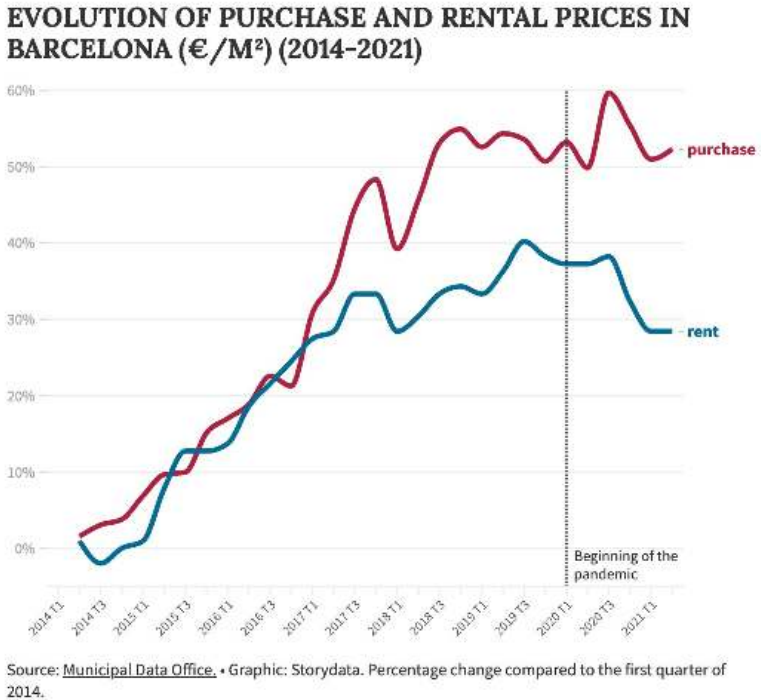
Set back major infrastructure behind a green buffer to prevent hard structures from causing wave reflection that scours away beach sand.

Establish a permeable green buffer between the ocean and built environment to absorb storm surges and manage runoff, preventing sediment wash-out.



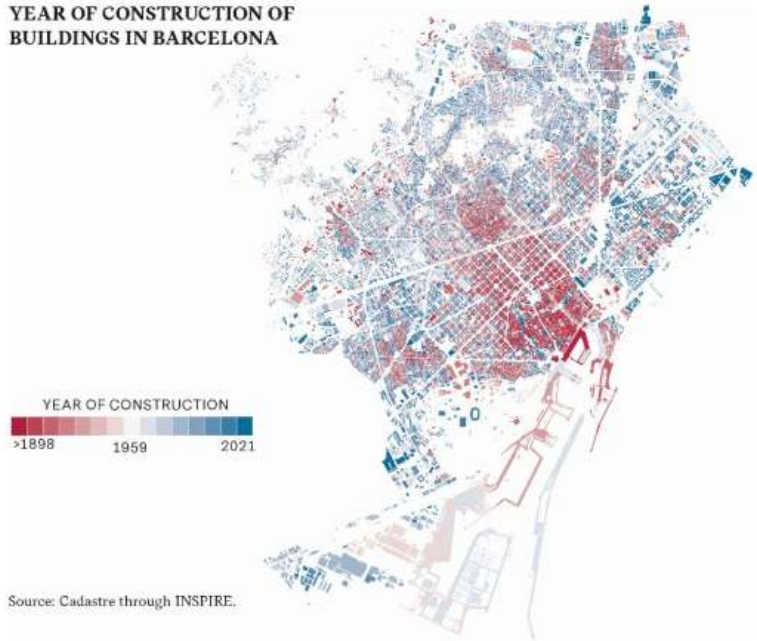
Site Analysis - Barcelona Housing Market & Crisis

- Access to decent housing defined as a **basic human right** in both Spanish Constitution and the Statue of Autonomy
- Average rent in Barcelona **increased 50%** between 2014 and 2021
- Shops and restaurants can **provide necessary profits** for the building while **keeping rent affordable**
- Spanish 2023 **Rent Cap Laws** protect renters



Site Analysis - Barcelona Building Energy Ratings

- Most buildings in the city **predate modern thermal insulation** efficiency standards
- Less than half of buildings (40.45%) have an energy efficiency certificate
    - Only 1.18% have an A or B rating
- 84.40% of buildings have received an E, F, or G, the **lowest possible ratings**
- Organizing the flats above to **where the heat rises** can boost energy efficiency



Site Analysis - Building Program

Public (~24% of interior area)

- Entry lobby (4%, 1 floor)
- Ground level retail & restaurant spaces (20%, 5 floor)

Semi-Public (~36% of interior area, 8% as transitional space)

- Open private work spaces for residents (4 floors, 16%)
- Residence Gym (1 floor, 4%)
- Rooftop park space (1 floor, 4%)

Private (~40% of interior space)

- Apartments (12 floors, 8% lost to semi-public transitional space)

Exterior spaces

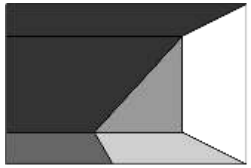
- Beach & sea access
  - Necessary to respond to Speculative Framework
- Ground level patio
- Site access to nearby rail
  - Existing Estacio St. Andria
  - Planned new station with Les Tres Xemeneies redevelopment

Project Statement

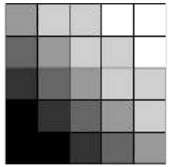
Our project goal is to analyze how manipulating exterior structure can influence interior spaces. Through studies of catenary arches and open forum precedents, we can test and create iterations based on relationships between curved geometry and structural systems.

Project Goals

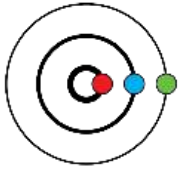
Maximize shaded indoor area to reduce heat during summer months



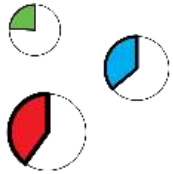
Maximized through Galapagos iterations and area totals



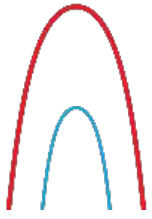
Interior areas proportional to set hierarchy



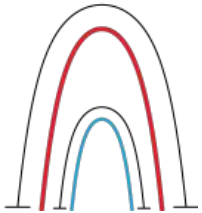
Measured by total areas of each defined space type



Hierarchy of main structural catenaries



Compared through ratios of measurements

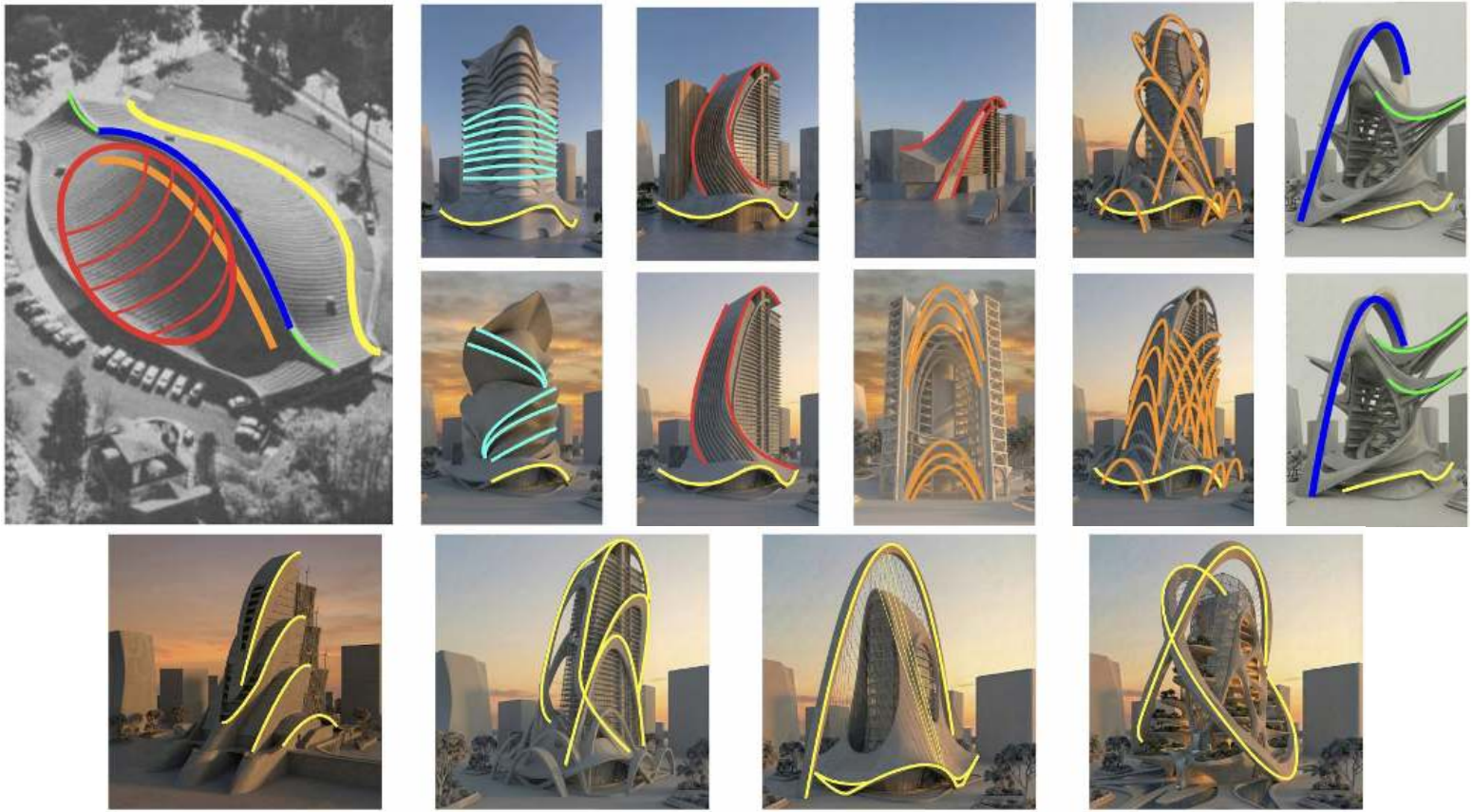




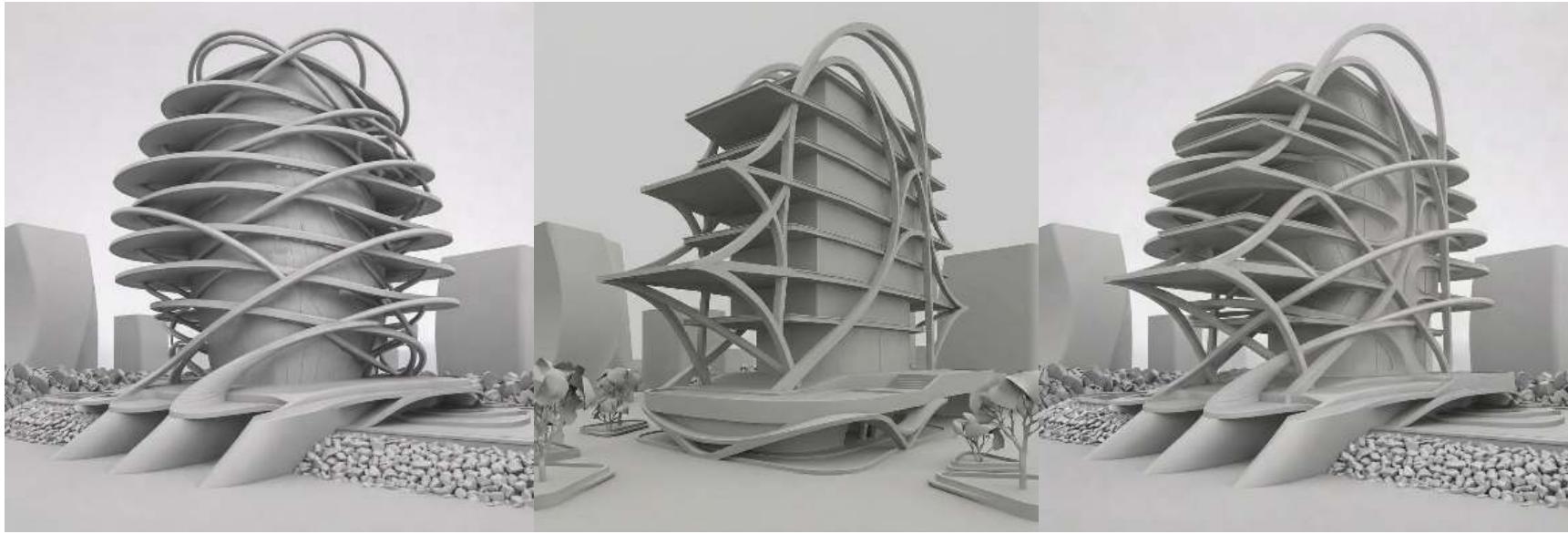
AI Iteration Investigation - Iteration Curve Analysis



AI Iteration Investigation - Iteration Curve Analysis

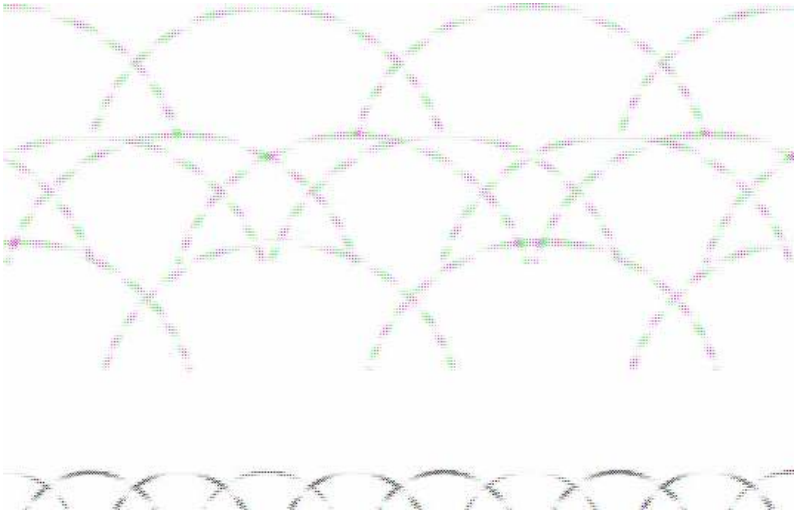


AI Iteration Investigation - Key Iterations

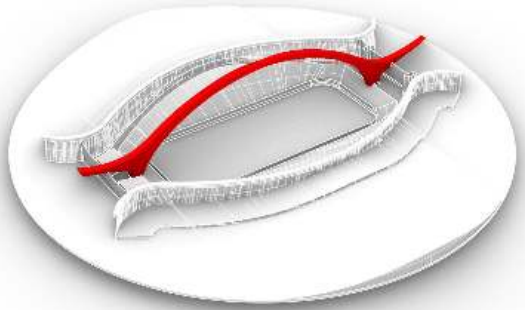


Design Rules - Fixed Parameter

Curve Hierarchy & Ratios

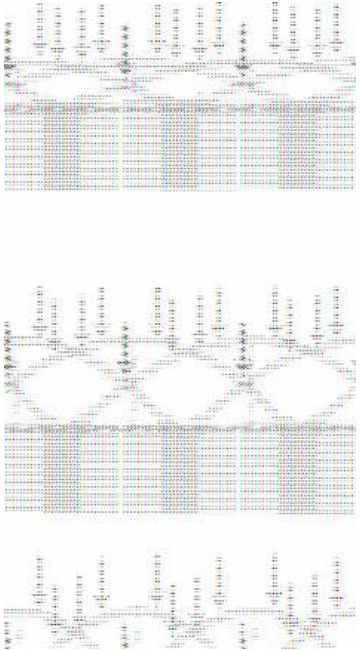


Catenary Structural Backbone

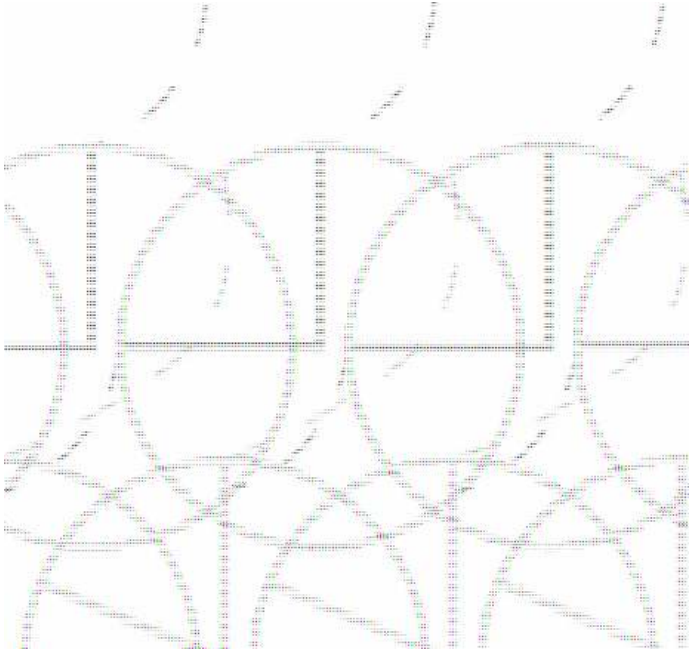


Design Rules - Variable Parameter

Catenary Spain & Height

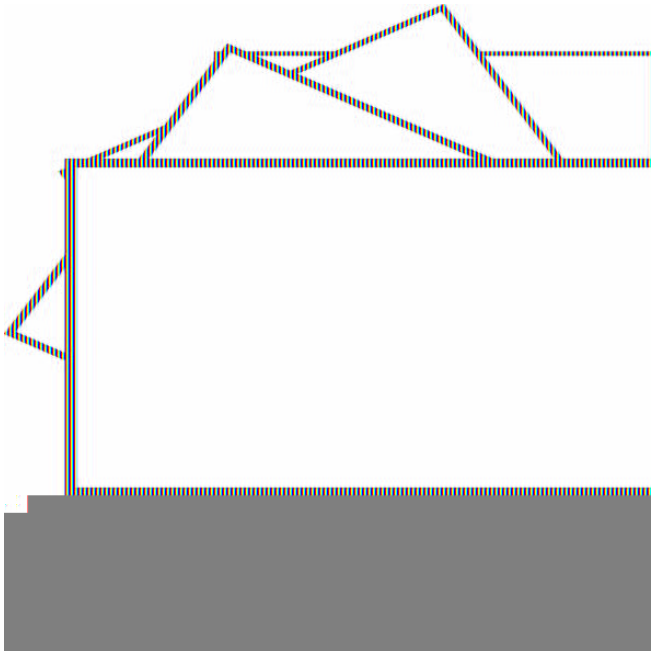


Angles Between Curves

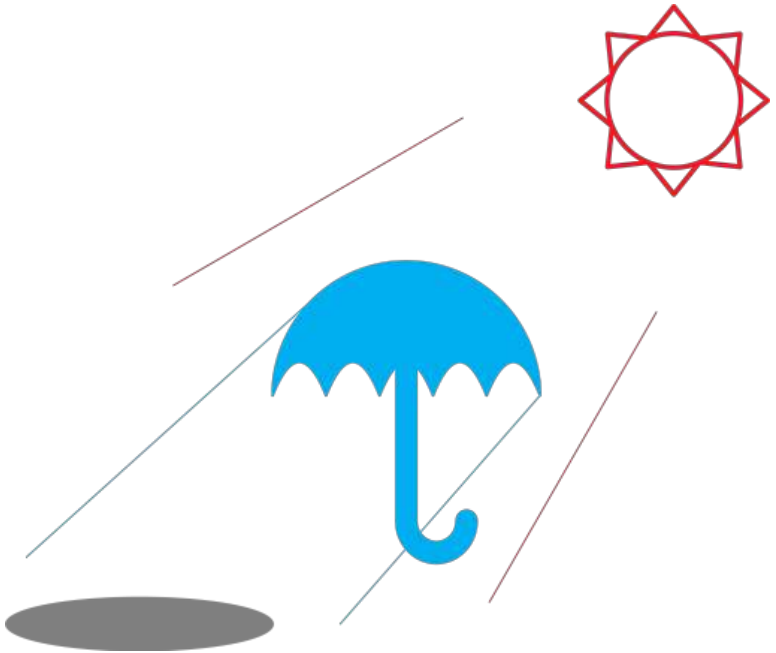


Design Rules - Variable Parameter

Floor Organization to Create Shading

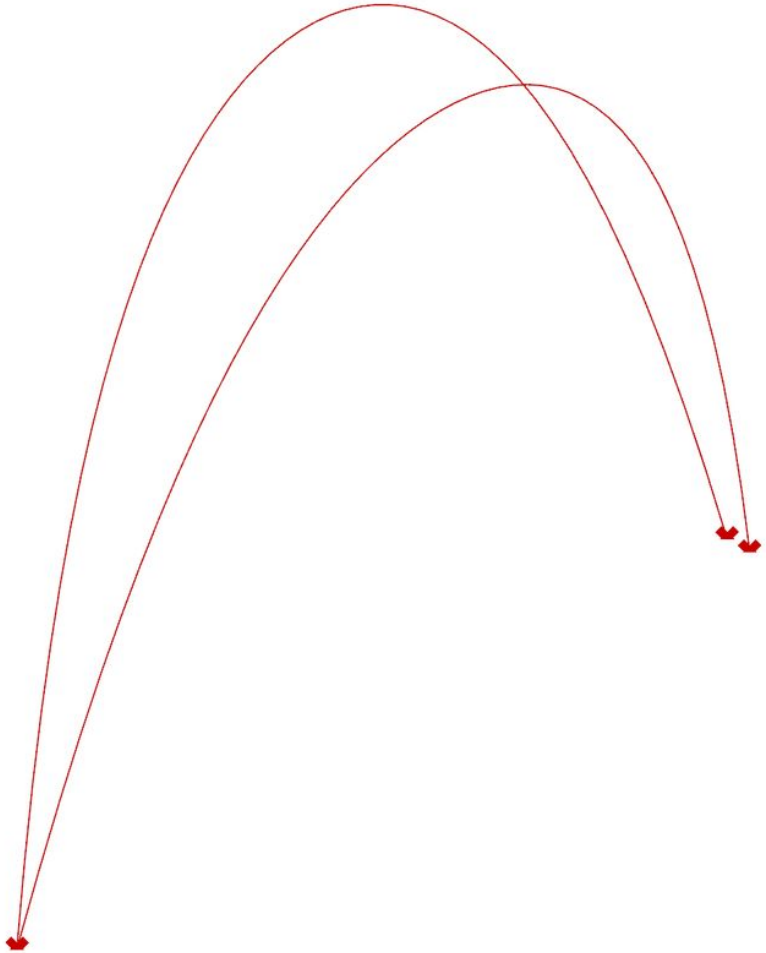


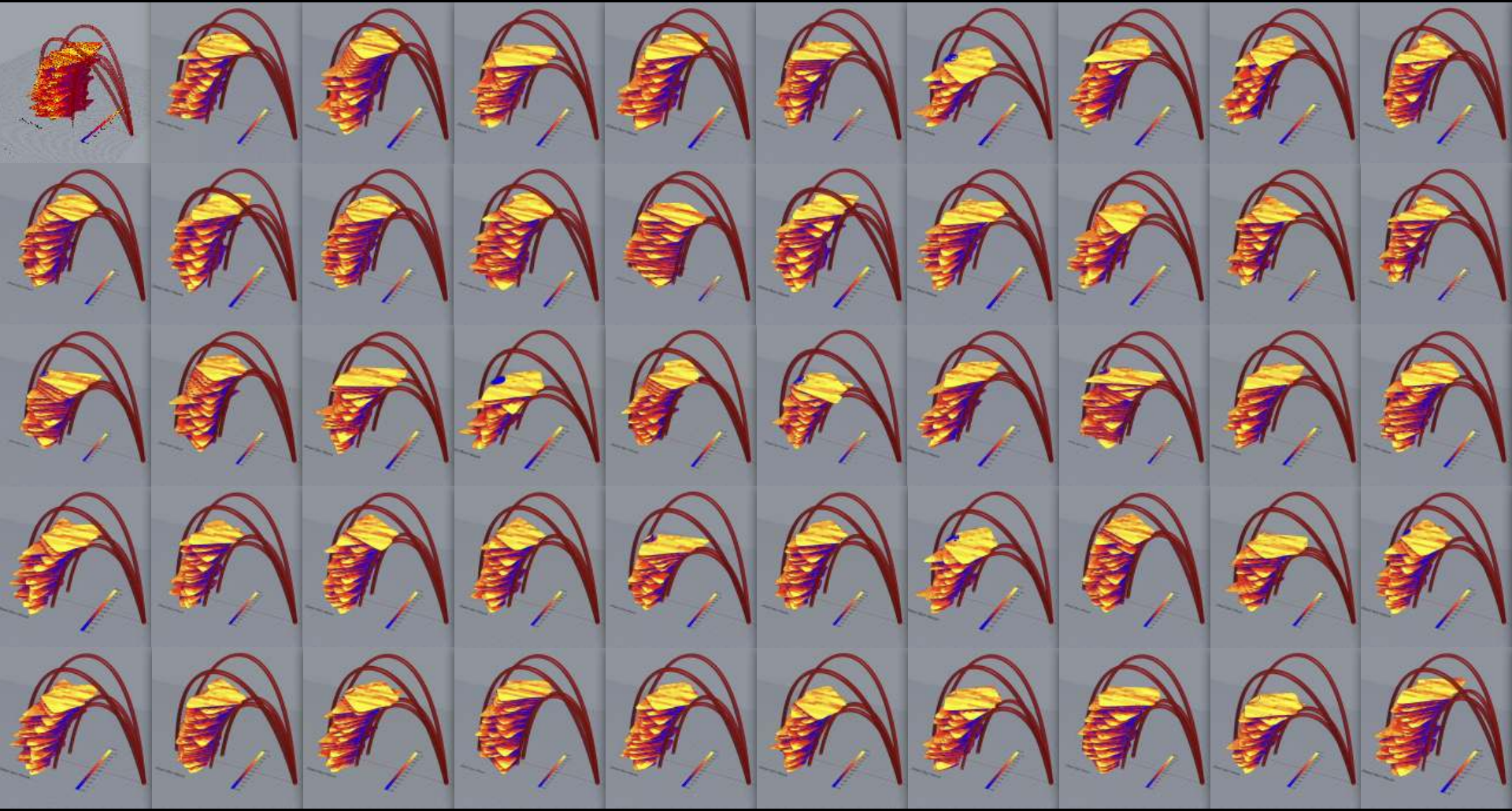
Indoor Shading from Coastal Sunlight

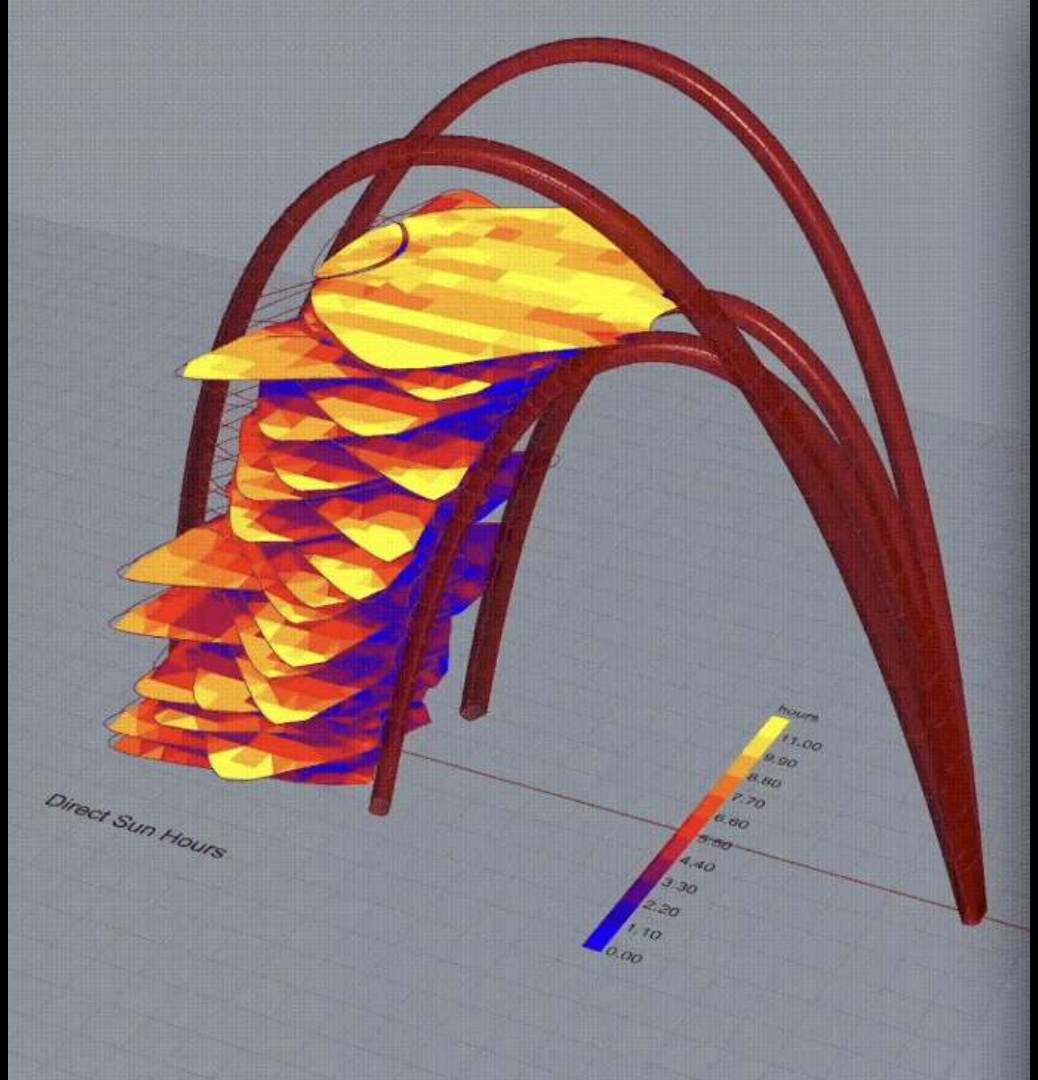


Design Process

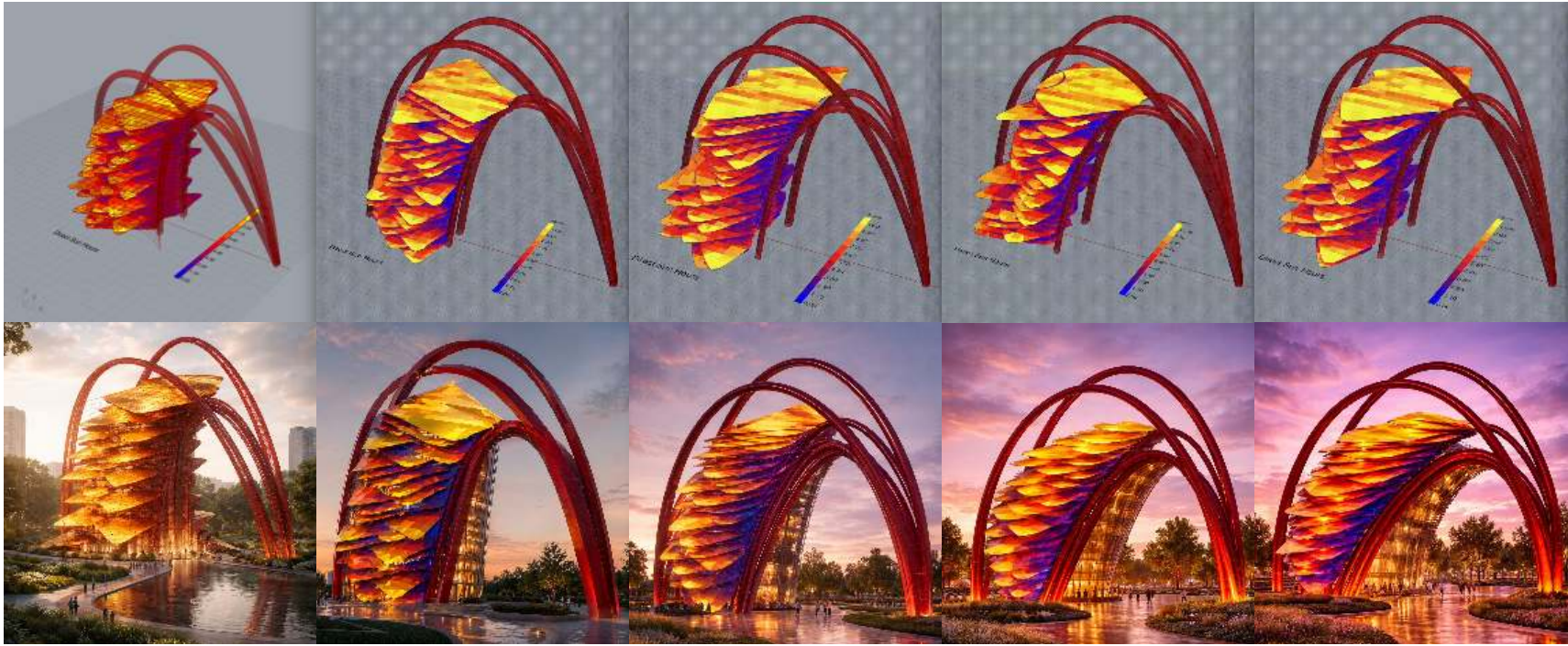
Protocol Animation

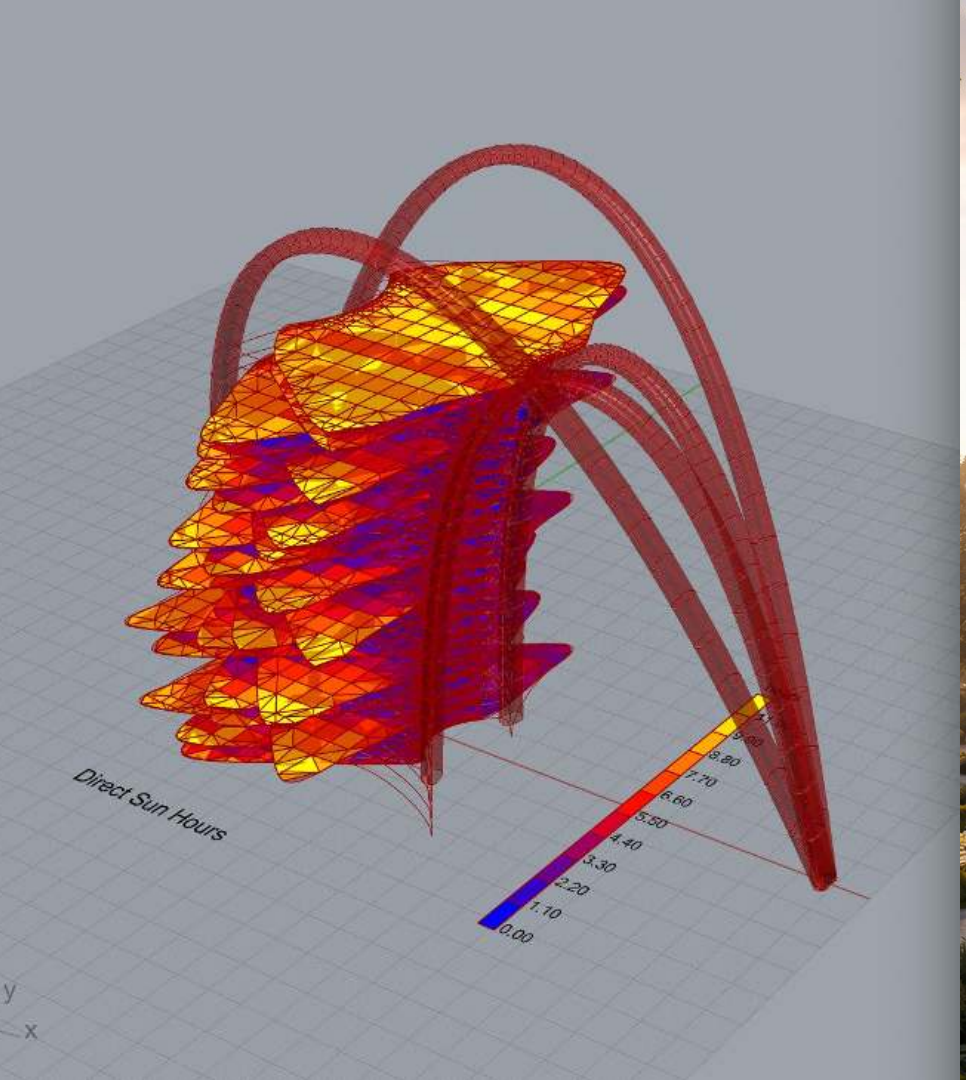




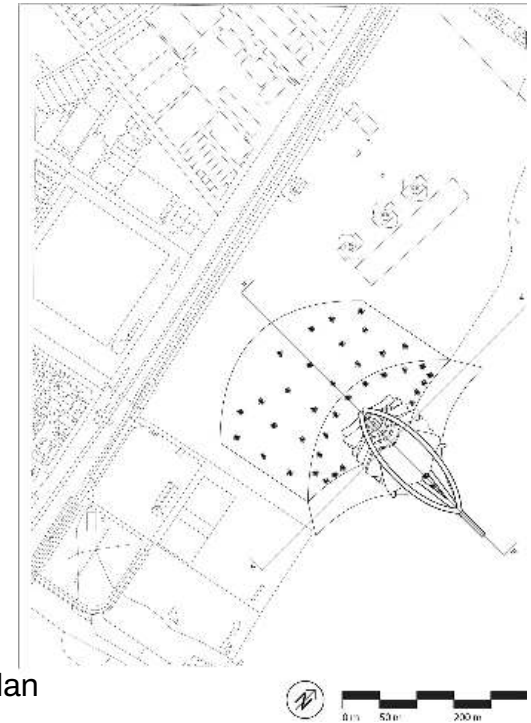

Direct Sun Hours
hours
11.00
9.90
8.80
7.70
6.60
5.50
4.40
3.30
2.20
1.10
0.00

Catalog of Iterations - Best Iterations

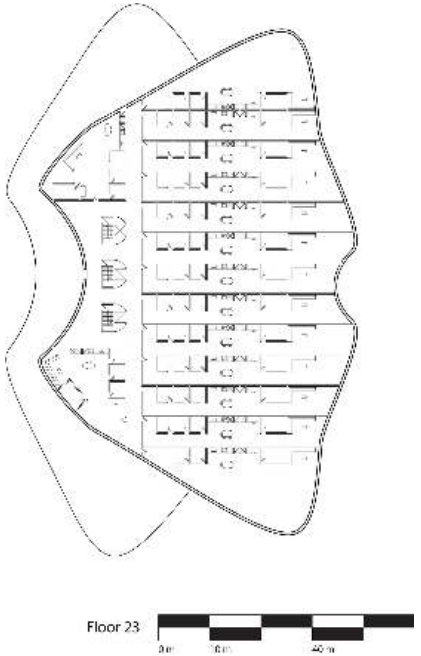



Direct Sun Hours
8.80
7.70
6.60
5.50
4.40
3.30
2.20
1.10
0.00
y
x

Selected Iteration - Site & Floor Plans

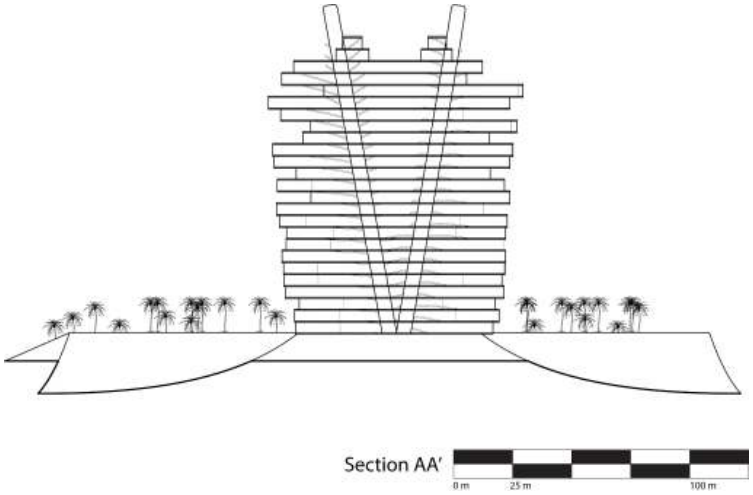


Site Plan

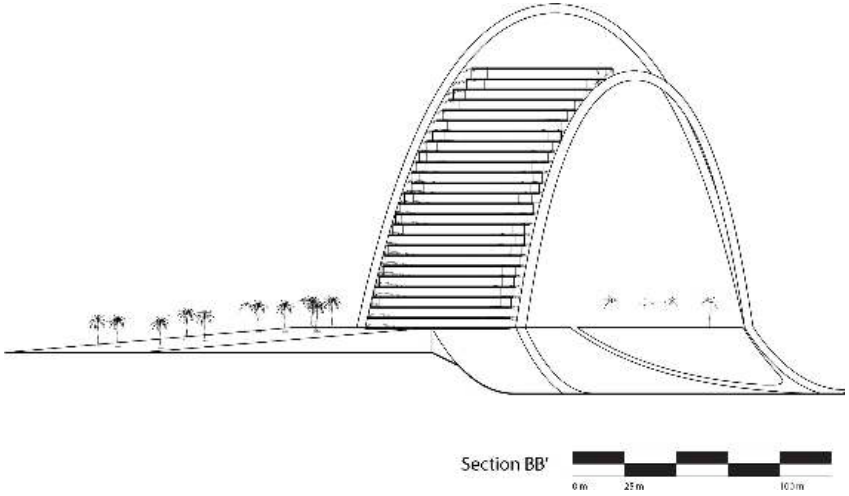


Floor Plan, upper residential floor

Selected Iteration - Section Views



Section AA', View from southeast



Section BB', View from southwest









Project Conclusion

- Successfully analyzed and found inspiration in a difficult precedent
- Strong influence from precedent incorporated into final design
- Strong supporting data for the speculative framework, but response to rising tides could have been better incorporated into tower's design
- Developed initial catalog of AI iterations with variety in design
- Thorough research of site at multiple scales (national, regional, city, and specific site)
- Successful in using Grasshopper to develop design and optimizing shaded area from Galapagos
- Growing pains using Grasshopper were able to be overcome
- Struggled with Galapagos to develop true catalog of iterations
- Final design responds well to project goals, but could have been better incorporated into the project
- Facade of final design could have been more detailed and developed, beyond a glass canopy with extruding balconies.

Overall, the project was well developed, however lacks connection to certain steps within our design process. Throughout the project, the design was adapted many times based on our precedent analysis and research which is where connection from previous work had been lost.

The next steps for the project would be studying the effectiveness of the base responding to deflecting waves away from the shoreline. Based on this research, redesign and development could be necessary.

